

Olist is a Brazilian marketplace that connects thousands of sellers with customers throughout Brazil. The platform collects information on orders, customers, products, payments, deliveries, and customer reviews.

The goal of the project is to analyze sales data and identify factors affecting revenue, delivery efficiency, and customer satisfaction. The project focuses on analyzing key performance indicators (KPIs) and answering business questions.

The dataset contains real data from the Olist platform for the years 2016–2018 and includes over 100,000 orders as well as information on customers, products, payments, deliveries, and customer reviews.

<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce?resource=download>

Business Questions:

1. Which product categories generate the highest revenue?
2. How have sales changed over time?
3. In which regions are sales the highest?
4. From which regions do the sellers generating the highest revenue come?
5. What is the average delivery time in each region?
6. Do delayed deliveries affect customer ratings?
7. Which product categories receive the lowest customer ratings?

Tables I plan to use:

- orders
- customers
- products
- product category name translation
- payments
- order items
- sellers
- reviews.

Sales - KPIs:

- Total Revenue (Sum of Price from the fact_orders table)
- Total Orders
- Average Order Value (AOV)

Delivery - KPIs:

- Average Delivery Time

- Late Delivery Rate (percentage of orders delivered after the estimated delivery date)
- Average Shipping Time

Customer Satisfaction - KPIs:

- Average Review Score
- Share of positive reviews (4-5)
- Share of negative reviews (1-2)

Tools Used:

- Python - EDA

<https://colab.research.google.com/drive/14w2USoLbNVtXWuVZjdfMw6WZp2AenUmC?usp=sharing>

- SQL - tables preparation

-- Tabela dim_products z nazwami kategorii po angielsku

```
SELECT
p.product_id,
ct.product_category_name_english
FROM products p
LEFT JOIN category_translation ct
ON p.product_category_name = ct.product_category_name
```

-- Tabela dim_customers

```
SELECT
customer_id,
customer_city,
customer_state
FROM customers
```

-- Tabela dim_payments (ze względu na powtarzające się order_id (płatności w ratach) payment_value zostało zsumowane dla każdego order_id)

```
SELECT
order_id,
SUM(payment_value)
FROM payments
GROUP BY order_id
```

-- Tabela dim_sellers

```
SELECT
seller_id,
seller_city,
seller_state
FROM sellers
```

-- Sprawdzenie zakresu dat zamówień

```
SELECT
MIN(order_purchase_timestamp),
MAX(order_purchase_timestamp)
FROM orders
```

-- Tabela fact_orders dla zamówień, których status to 'delivered', podana jest data dostarczenia zamówienia (żeby wyliczyć różnicę między datą przewidywaną a rzeczywistą) i datach ograniczonych do pełnych miesięcy

-- Żeby zbudować schemat gwiazdy, połączono tabelę orders i order_items.

```
SELECT *  
FROM orders o  
JOIN order_items oi  
ON o.order_id = oi.order_id  
WHERE o.order_status = "delivered" AND o.order_delivered_customer_date IS NOT NULL  
AND o.order_purchase_timestamp BETWEEN '2016-10-01' AND '2018-09-30'
```

-- Sprawdzenie, czy każde zamówienie ma tylko jedną recenzję, żeby model gwiazdy działał poprawnie

```
SELECT  
order_id,  
COUNT(*) AS order_id  
FROM reviews  
GROUP BY order_id
```

-- Ze względu na to, że niektóre zamówienia miały więcej niż jedną recenzję, do analizy wybrano tylko najnowszą recenzję, tak by pojawiały się jedynie unikalne order_id, by schemat gwiazdy działał poprawnie

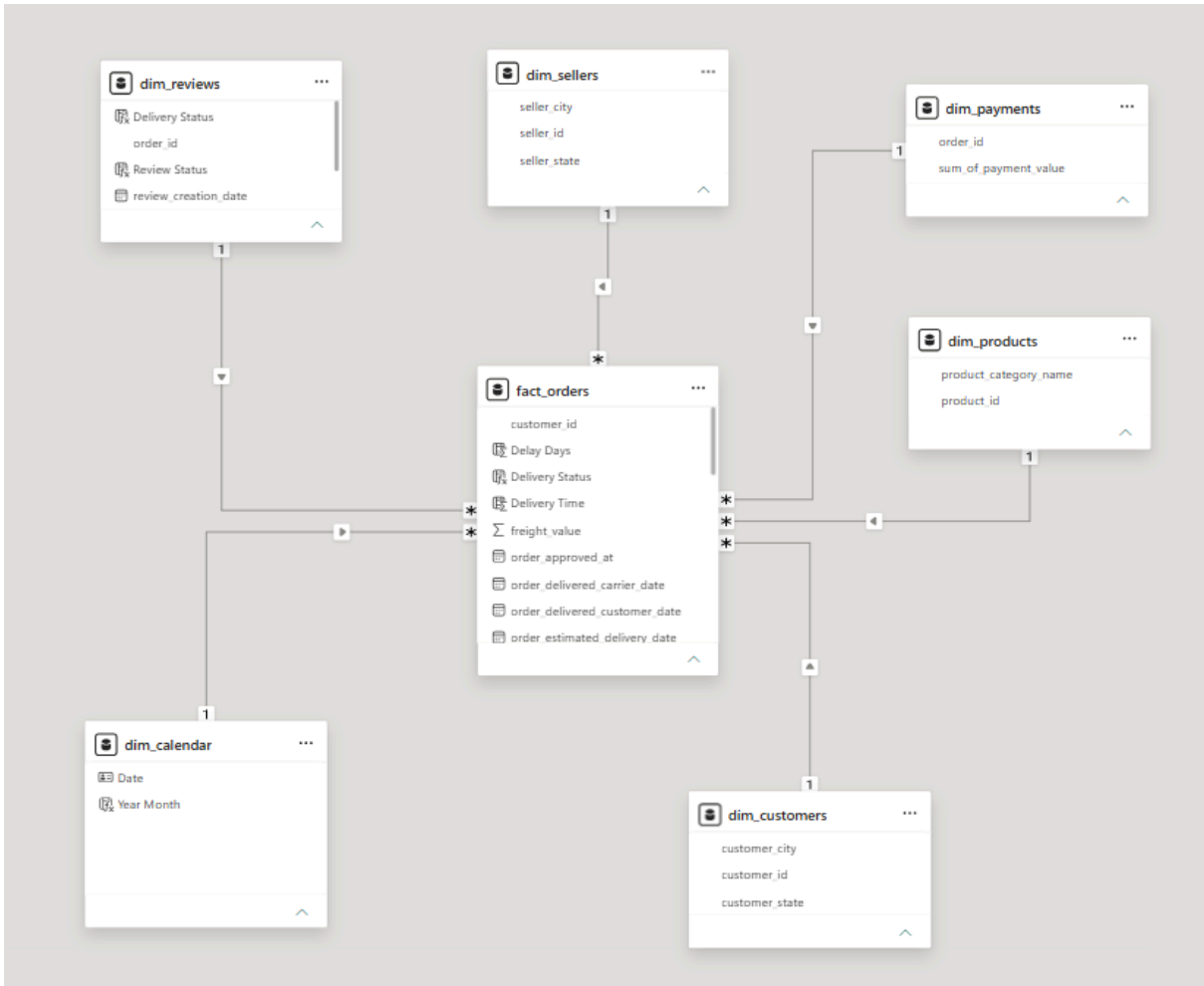
```
WITH ranked_reviews AS  
(SELECT  
order_id,  
review_id,  
review_score,  
review_creation_date,  
ROW_NUMBER() OVER (PARTITION BY order_id ORDER BY review_creation_date DESC) AS rn  
FROM reviews)  
SELECT  
order_id,  
review_id,  
review_score,  
review_creation_date  
FROM ranked_reviews  
WHERE rn = 1
```

The following tables were created based on the source tables:

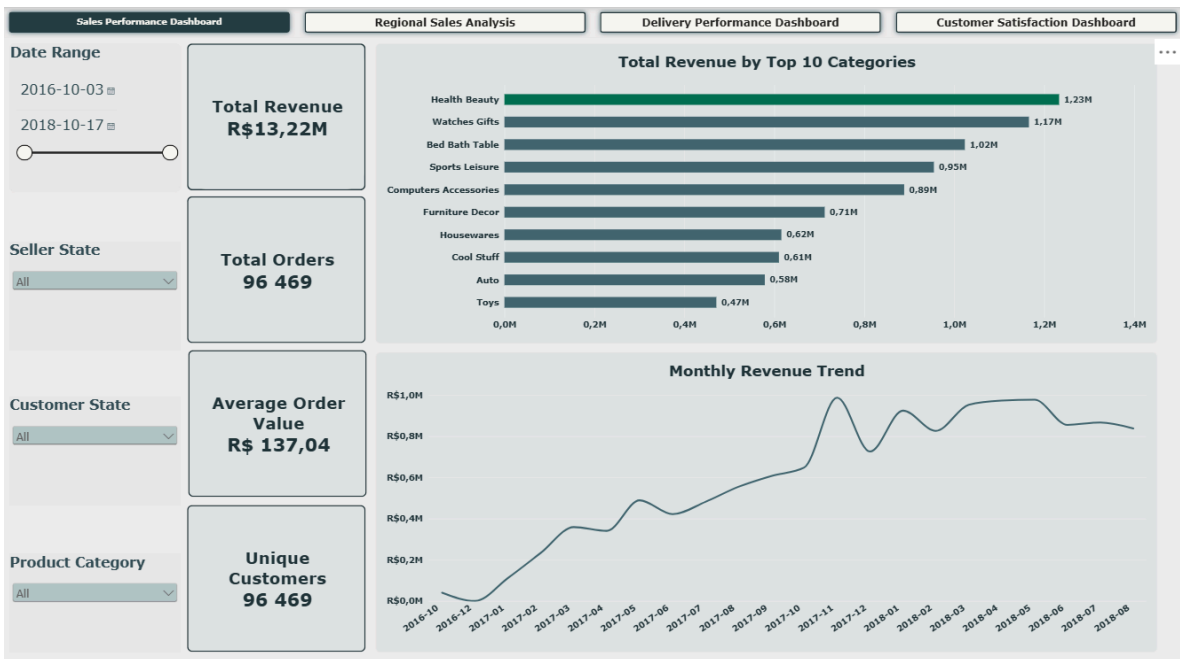
- fact_orders
- dim_sellers
- dim_products
- dim_reviews
- dim_payments
- dim_customers
- dim_calendar

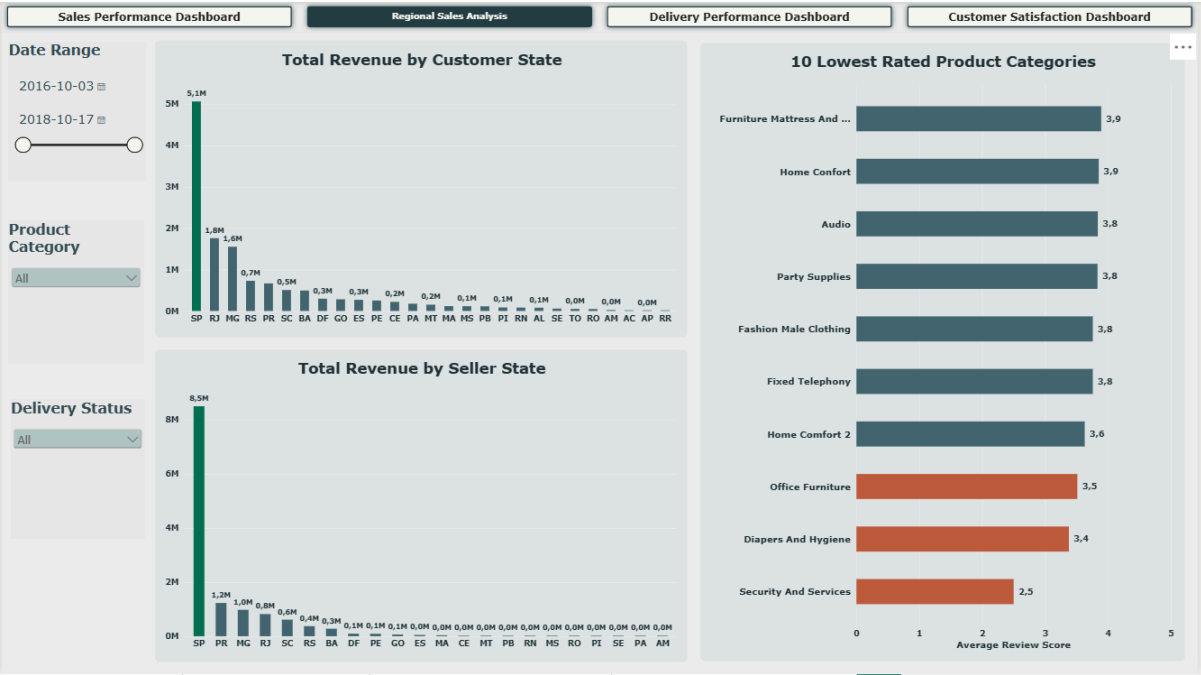
- **Power BI**

Star schema:



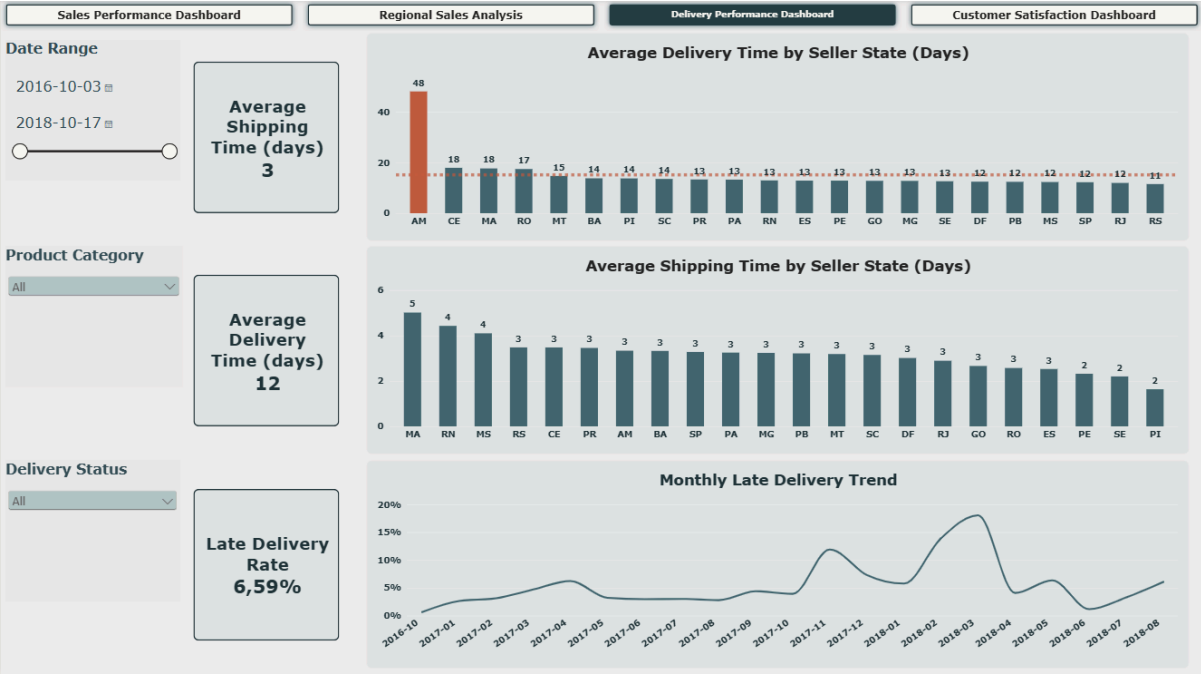
Visualizations:





10 Lowest Rated Product Categories

Product Category	Average Review Score
Furniture Mattress And ...	3.9
Home Comfort	3.9
Audio	3.8
Party Supplies	3.8
Fashion Male Clothing	3.8
Fixed Telephony	3.8
Home Comfort 2	3.6
Office Furniture	3.5
Diapers And Hygiene	3.4
Security And Services	2.5



Average Delivery Time (days)
12

Late Delivery Rate
6,59%

